

UCT201: LINEAR ALGEBRA

L	T	P	Cr
3	1	0	3.5

Course Objectives: The Linear algebra course provide the in-depth knowledge about the mathematical formulation building for real world problems in engineering and technology. Students will be able to learn the vector space and linear combination, orthogonality and projections, spanning of vector space and their fundamental relationship with computer science.

Introduction to Matrices and Determinants: Solution of Linear Equations; Cramer's rule; Inverse matrix.

Vectors and linear combinations: Rank of a matrix; Gaussian elimination; LU decomposition; Solving systems of linear equations using the tools of matrices.

Vector space; Dimension; Basis: Orthogonality; Projections; Gram-Schmidt orthogonalization and QR decomposition.

Eigenvalues and Eigenvectors: Positive definite matrices; Linear transformations; Hermitian and unitary matrices.

Singular value decomposition and Principal component analysis: Introduction to their applications in Image Processing and Machine Learning.

Course Learning Outcomes (CLOs)/ Course Objectives (COs):

The students will be able to:

1. Apply basic concept for the solution of a linear system, $AX=B$ and find all solutions by choosing an effective method such as Cramer's rule; Inverse matrix and Gaussian elimination.
2. Learn about the vector space and linear combination, orthogonality and projections, spanning of vector space and their fundamental relationship with computer science.
3. Find the Gram-Schmidt orthogonalization of a matrix; QR decomposition, determine the rank, determinant, eigenvalues and eigenvectors, diagonalization, and different factorizations of a matrix.
4. Identify special properties of a matrix, such as symmetric of Hermitian, positive definite, etc., and use this information to facilitate the calculation of matrix characteristics.
5. Use the singular value decomposition and principal component analysis in the practical applications like medical image denoising and machine learning.

Text Books:

1. Seymour Lipschutz, Schaum's Outline of Linear Algebra, McGraw Hill Education, 3rd edition, 2017.
2. Gilbert Strang, Introduction to Linear Algebra, Wellesley-Cambridge Press, 5th Edition, 2016.

Reference Books:

1. David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and Its Applications Loose Leaf, Pearson College Div, 5th edition, 2015.
2. Kuldeep Singh, Linear Algebra: Step by Step, Oxford University Press, 1st Edition, 2013.